Chapter 8, Problem 61P

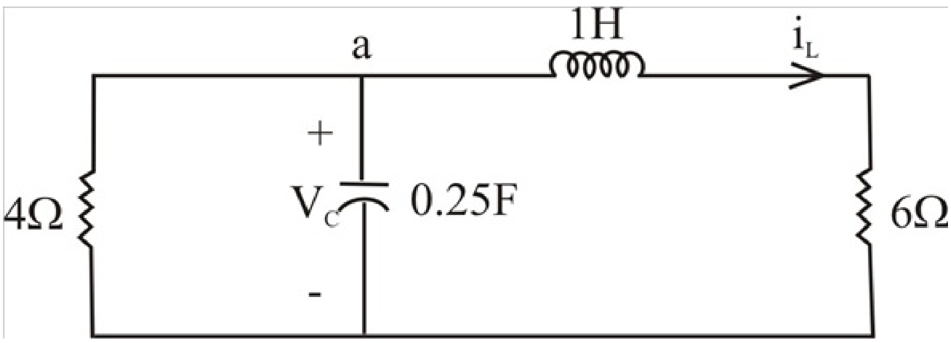
Problem

For the circuit in Prob. 8.5, find *i* and *v* for *t* > 0.

Step-by-step solution

Step 1 of 4

For t > 0, we obtain the natural response by ∞ considering the circuit below.



Step 2 of 4

At node a,

$$\frac{V_C}{4} + 0.25\frac{dV_C}{dt} + i_L = 0 \quad \rightarrow (1)$$

$$V_C = 1\frac{di_L}{dt} + 6i_L \quad \rightarrow (2)$$

Combining (1) & (2),

$$\left(\frac{1}{4}\right)\frac{di_L}{dt} + \frac{6}{4}i_L + 0.25\frac{d^2i_L}{dt^2} + \frac{6}{4}\frac{di_L}{dt} + i_L = 0$$

$$\frac{d^2i_L}{dt^2} + \frac{7\,di_L}{dt} + 10i_L = 0$$

$$S^2 + 7S + 10 = 0$$

$$\Rightarrow \quad (S+2)(S+5) \quad Or \quad S_{1,2} = -2,\ -5$$

Step 3 of 4

Thus, $i_L(t) \quad = \quad i_L(\infty) + [Ae^{-2t} + Be^{-5t}]$

Where,

$i_L(\infty)$ represent final inductor current $= \dfrac{4(4)}{(4+6)} = 1.6$

$$i_L(t) = 1.6 + [Ae^{-2t} + Be^{-5t}] \quad \& \quad i_L(0)$$
$$= 1.6 + [A+B] \quad Or \quad -1.6 = A + B \rightarrow (3)$$

$$\frac{di_L}{dt} = [-2Ae^{-2t} - 5Be^{-5t}]$$

And,

$$\frac{di_L(0)}{dt} = 0 = -2A - 5B \quad Or \quad A = -2.5B \rightarrow (4)$$

Step 4 of 4

For (3) & (4),

$$A = \frac{-8}{3} \quad \& \quad B = \frac{16}{5}$$

$$i_L(t) = 1.6 + \left[-\left(\frac{8}{3}\right)e^{-2t} + \left(\frac{16}{5}\right)e^{-5t}\right]$$

$$V(t) = 6i_L(t) = \{9.6 + [-16e^{-2t} + 6.4e^{-5t}]\}\ V$$

$$V_C = \frac{di_L}{dt} + 6i_L = \left[\left(\frac{16}{3}\right)e^{-2t} - \frac{16}{3}e^{-5t}\right]$$
$$+ \{9.6 + [-16e^{-2t} + 6.4e^{-5t}]\}$$

$$V_C = \left\{9.6 + \left[-\left(\frac{32}{3}\right)e^{-2t} + 1.0667e^{-5t}\right]\right\}$$

$$i(t) = \frac{V_C 4}{} = \left\{2.4 + \left[-2.667e^{-2t} + 0.2667e^{-5t}\right]\right\} A$$

[Comments (1)]

Anonymous

I believe B should be 16/15, not 16/5